

Open CV Library

→ Basic :

Reading Images and Videos

Image Transformation

Drawing Shapes

→ Faces :

Face Detection

Face Recognition

Deep CV

→ Advanced :

Color Spaces

Bitwise Operations

Masking

Histogram Computation

Edge Detection

→ Reading

~~★~~ import cv2 as cv

img = cv.imread('path')

cv.imshow(name, img)

cv.waitKey(0) → *click any key to close*

Video

capture = cv.VideoCapture('path')

while True :

if True, frame = capture.read()

if not is True :

break

cv.imshow('name', frame)

if cv.waitKey(20) & 0xFF == ord('q')

break

Capture Image C)

CV.destroyAllWindows()

Essential Functions:-

import cv2 as cv

Convert to grayscale

gray = cv.cvtColor(img, cv.COLOR_BGR2GRAY)

Blur

blur = cv.GaussianBlur(img, (7, 7), cv.BORDER_DEFAULT)

Edge Detection

canny = cv.Canny(blur, 125, 175)

Dilation

dilation = cv.dilate(canny, (7, 7), iterations=1)

Erosion

eroded = cv.erode(dilation, (7, 7), iterations=1)

cv.imshow('Canny Edges', canny)

cv.waitKey(0)

drawing shape and write text

import cv2 as cv

import numpy as np

black image of 500x500 Pixels

blank = np.zeros((500, 500, 3), dtype='uint8')

Color Part of image b2 Green

blank[200:300, 300:400] = 0, 255, 0

Width Length R G B

Rectangle

cv.rectangle(blank, (0, 0), (250, 250),
(0, 255, 0), thickness=2)

(image, Start Point, End Point,

Color, thickness)

Circle

cv.Circle(blank, (250, 250), 40,
(0, 0, 255), thickness=1)

(image, center, radius, color, thickness)

Text

```
cv.putText(blank, "Text", (0, 225),
cv.FONT_HERSHEY_TRIPLEX, 1.0,
(0, 252, 0), 2)
```

```
cv.imshow('Drawn', blank)
cv.waitKey(0)
```

Resizing, Cropping, Flipping

```
import cv2 as cv
img = cv.imread('path')
```

Resized

```
resized = cv.resize(img, (500, 500),
interpolation = cv.INTER_AREA)
```

Cropping

```
cropped = img[50:200, 200:400]
```

```
flip_horiz = cv.flip(img, 1)
```

```
flip_vert = cv.flip(img, 0)
```

```
flip_both = cv.flip(img, -1)
```

Split the 3 Channel of image

```
b, g, r = cv.split(img)
```

merge the 3 Channel

```
merged = cv.merge([b, g, r])
```


Masking

→ Cropping the image as a circle

```
import cv2 as cv
import numpy as np
img = cv.imread('path')
blank = np.zeros(img.shape[:2],
                  dtype='uint8')
mask = cv.circle(blank, (img.shape[0]//2,
                          img.shape[1]//2),
                  100, 255, -1)
```

```
masked_img = cv.bitwise_and(
    img, img, mask=mask)
```


→ Face detection using Haar Cascade

```
import cv2 as cv
img = cv.imread('Path')
gray = cv.cvtColor(img, cv.COLOR_
BGR2GRAY)
```

```
haar_cascade = cv.CascadeClassifier
('haar-face.xml')
```

```
faces_rect = haar_cascade.detectMultiScale
(gray, scaleFactor=1.1, minNeighbors=3)
```

```
Print('number of Faces found: %len
faces_rect')
```

~~## draw rectangle on faces~~

```
for (x, y, w, h) in faces_rect:
    cv.rectangle(img, (x, y), (x+w, y+h),
    (0, 255, 0), thickness=2)
```

```
cv.imshow('detected Faces', img)
```

~~cv.waitKey~~

```
cv.waitKey(0)
```

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